

1	Course Name:
	Advanced Computer Science
2	Course Code:
	IDB40303
3	Year and Semester Offered:
	Year 2 Semester 2
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	NIL
6	Course Synopsis:
	Computer science changes the way people think solve problems and understand the world. Computer science describes about “how to” knowledge. This course discusses the computer science principles and programming practices which elaborates on the science of computing.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Inquire knowledge of the computer science, programming and theory of computation. 2. Demonstrate in group and present the overall computer science concept through the layers: information, programming, applications, operating systems and communication. 3. Develop solutions to computer science and ethical operational problems by selecting and developing an algorithm based on the object-oriented paradigm.
8	Content Outline:
	Topic
	1. Introduction to Computer Science
	2. The Operating Systems and Communication Layers
	3. The Information Layer: Data Representation
	4. The Hardware Layer: Computer Structure
	5. The Applications Layer
	6. The Programming Layer
9	References:
	Behrouz Forouzan, Foundation of Computer Science, Cengage, 2018